

Name: _____



Degrees of accuracy

Task A: Understanding degrees of accuracy

1) Complete the table to show the most appropriate degree of accuracy:

	Nearest integer	Nearest ten	Nearest hundred	Nearest thousand
The number of baked beans in a tin				
The height at which a plane flies				
Number of seats in 5 cinema screens				
The number of people living in a street				
Number of pages in a book				
Number of buckets of water to fill a paddling pool				
The number of grains of rice in 1 kg				

2) Complete the table to show which degree of accuracy is most likely to have been used:

	1 s.f.	2 s.f.	3 s.f.	4 s.f.
The population of a town is 12 000				
The length of a plank of wood is 6.24 m				
The width of a blade of grass is 1.6 mm				
The attendance at a pop concert is 30 000				
62% of UK homes own a pet				
The width of a human hair is 0.009 cm				
The distance from Earth to the Moon is 384 400 km				

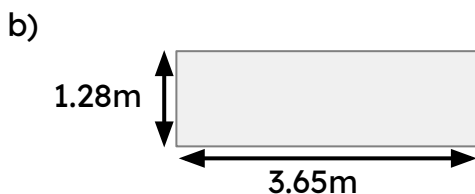
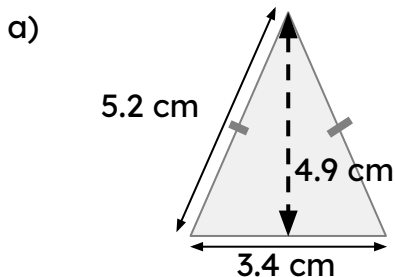
3) Rewrite the story below rounding each of the numbers to an appropriate degree of accuracy.

Sofia belongs to a supporters club for her favourite football team. They decide to organise coaches to watch an away match. 19 coaches of 52 people set off for the match. The coach travels 262.504 km to the match which takes them 3 hours, 53 minutes and 15 seconds. Whilst at the match Sofia and her mum buy some lunch which costs £18.97. During the match there was an announcement that the attendance at the match is 51 699. The average speed of the coach on the way home is 54.89 mph. And they arrive home at 23:58:42.

Task B: Answers to an appropriate degree of accuracy

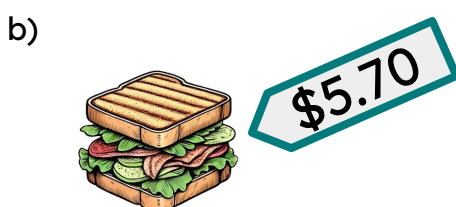
In each of the following questions give your answers to an appropriate degree of accuracy.

1) Calculate the area and perimeter of the following:



2) Convert the following from dollars into pounds:

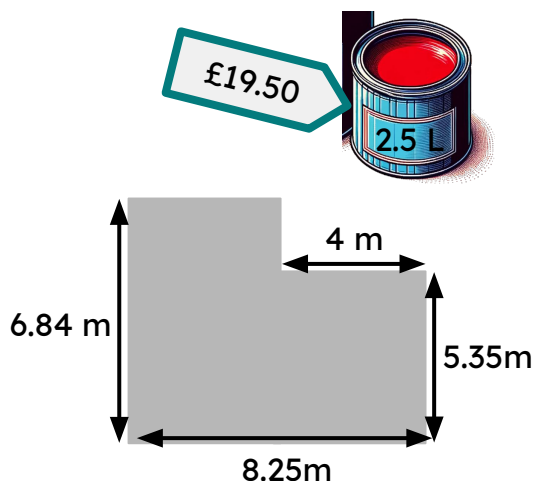
Using the exchange rate £1 : \$1.27



3) Answer the following. At each stage give your answer to an appropriate degree of accuracy.

How much will it cost to paint this wall?

1 tin of paint covers 25 m^2 .



4) Use your calculator to answer the following:

a)
$$\frac{\sqrt{22} + 3.8}{380 - 0.54}$$

b)
$$\frac{1.28 + (11.05)^2}{0.07 \times 3.92}$$

